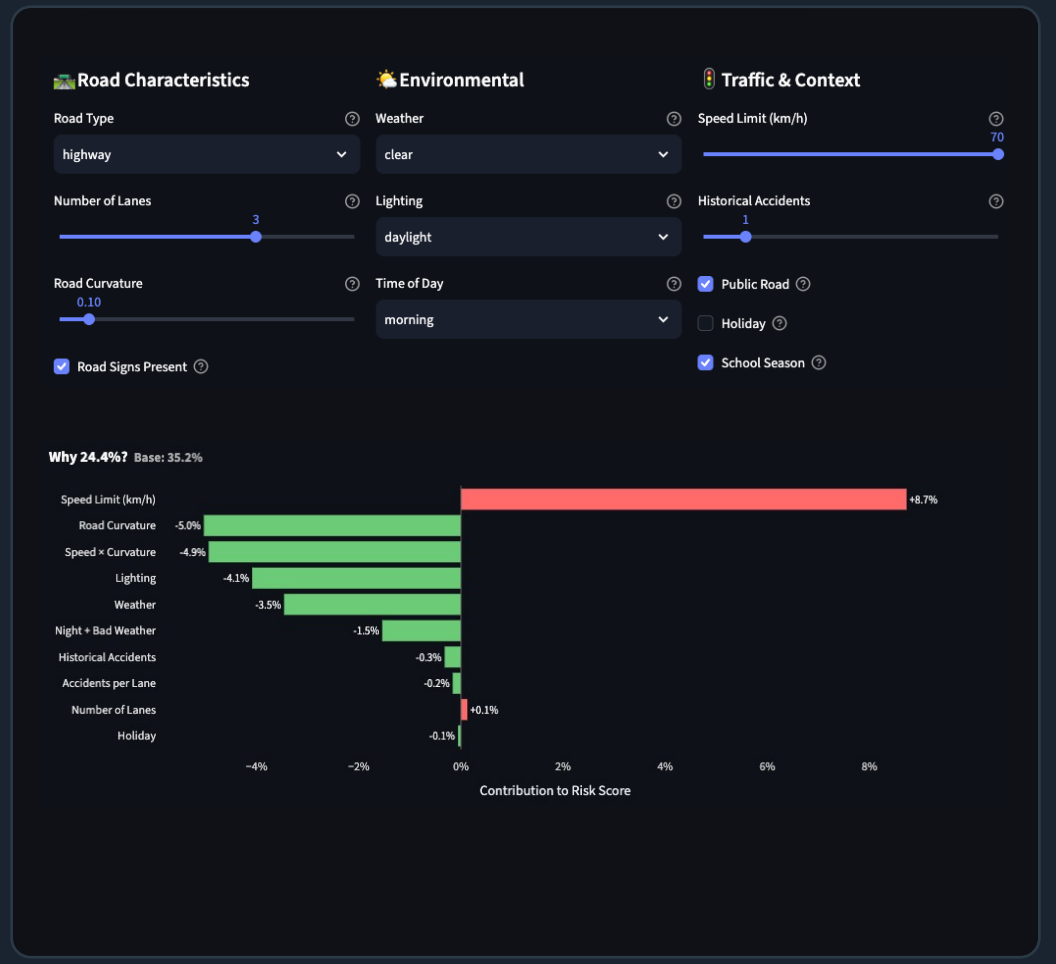
SSD 2026 · PROJECT P21

Safe-Route

An explainable shortlist for road safety checks

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Working prototype: Streamlit app

Which roads should engineers inspect first?

Time and budgets are limited. Safe-Route suggests where to look first.



12 sites · **time for 3 visits**

Three terms used in this talk

Each one is explained again when it comes up.

01

Risk score

A number from 0 to 1. Higher means the model rates the conditions as riskier.

02

Explanation

Which conditions pushed the score up or down, so a person can check it.

03

Practice data

Synthetic examples, not real New Zealand crash records.

From road conditions to an explained score

The tool suggests. An engineer still decides.

1

Describe the road

Road type, curve, speed limit, lighting, weather and more.

2

Get a score

A Random Forest model rates the conditions from 0 to 1.

3

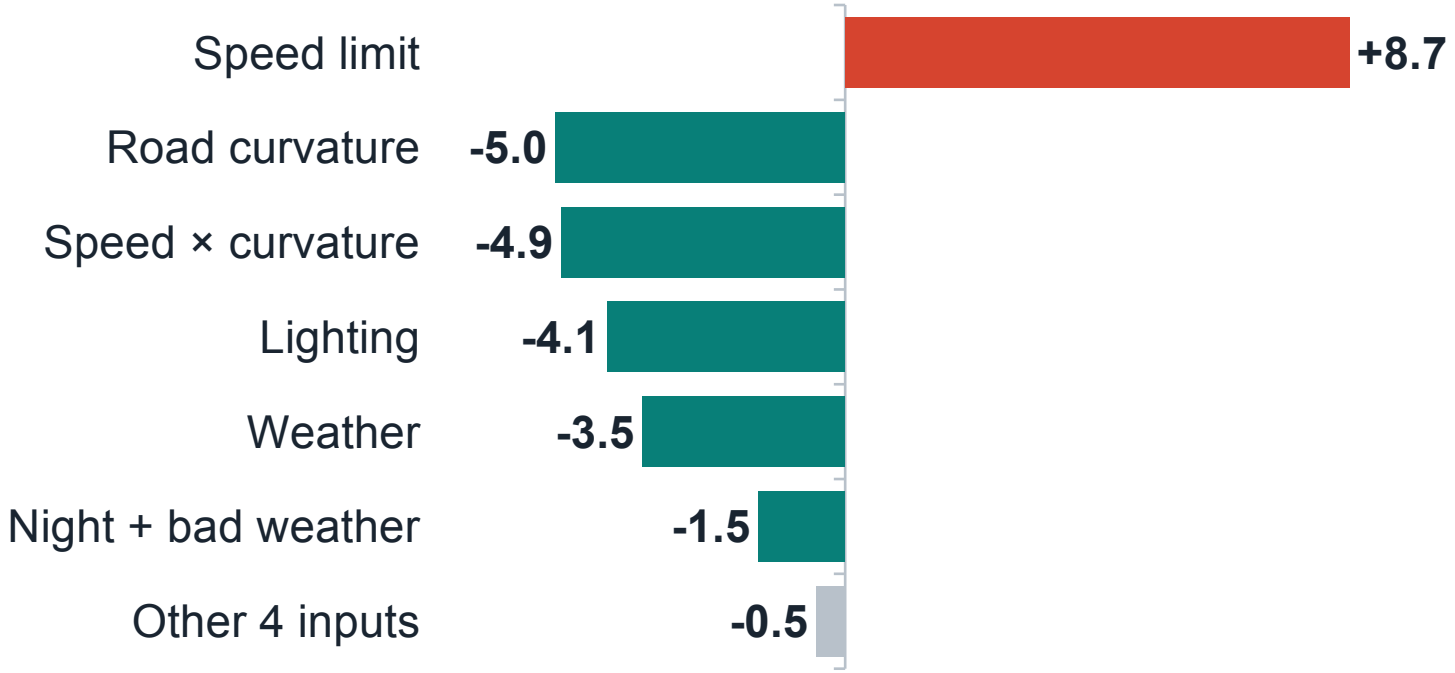
See why

SHAP shows which conditions raised or lowered the score.

Dataset: about 518,000 synthetic examples (Kaggle Playground Series S5E10). Model: 200 decision trees, 15 inputs.

What raised and lowered this score?

Example: highway, 70 km/h, daylight, clear weather, gentle curve.



Contribution to the score in points (app's 0 to 100 display), from TreeSHAP.

35.2 → 24.4

Average score, then this road once the bars are added. Low: below 75% of practice examples.

Raises the score

70 km/h speed limit

Lowers the score

gentle curve, daylight, clear weather

⚠ **The app shows 24.4%. That is a score, not a one-in-four chance of a crash.**

The model explains 88% of the score's variation

Measured on practice examples it had never seen. Real roads still need testing.

88%

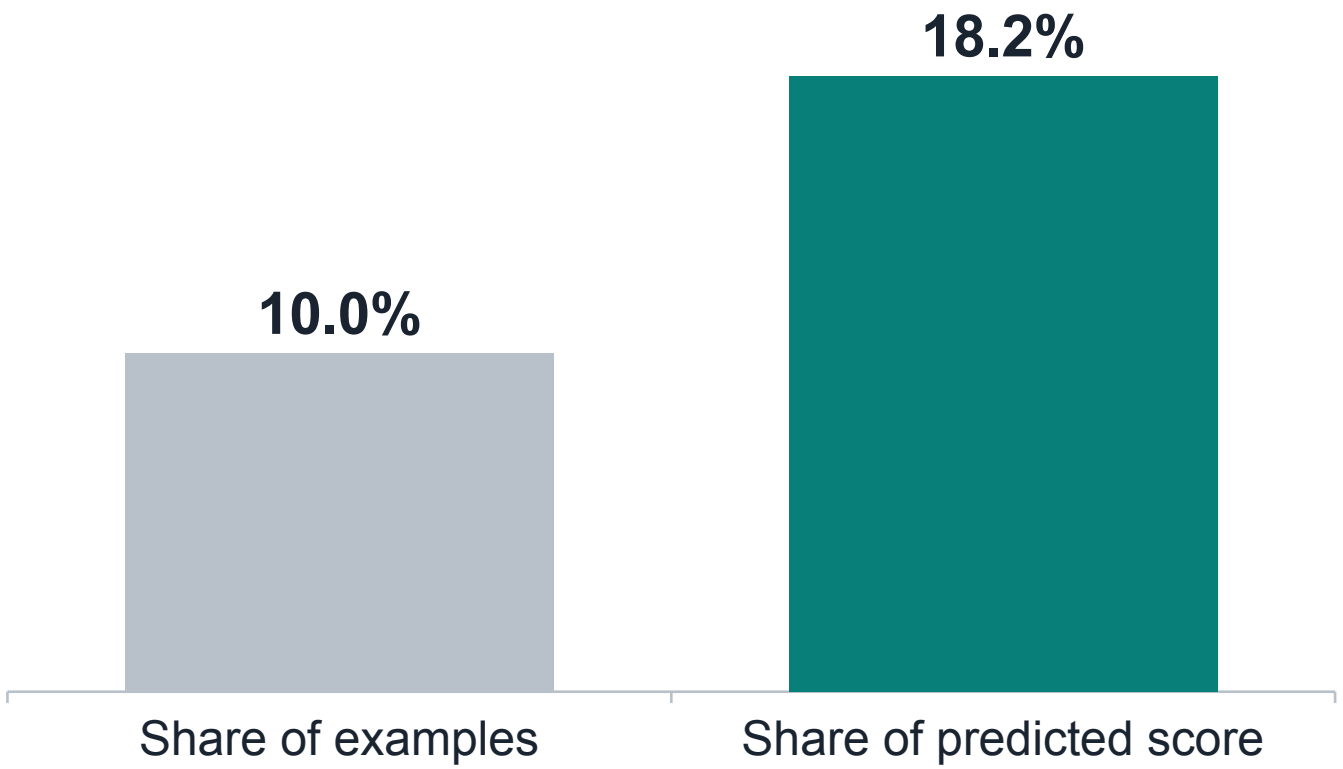
of the variation in the practice score, on one in five examples held back from training.

R^2 0.8835
Average error 0.044 on a 0 to 1 scale

Slightly weaker at night: 85.7% in daylight, 84.6% at night.

The top 10% hold 18.2% of the predicted score

Sort every practice example by its score, then look at the highest tenth.



1.82×

their fair share of the predicted score

Measured on the model's own scores. Whether it finds real problem sites is the next test.

Three limits before real-world use

Built for planning roads, not judging drivers. No personal data is used.

1	**Not a real crash probability**	The model learned from synthetic practice data.
2	**Not tested in other places or years**	The data has no locations or dates.
3	**Not proof that a fix works**	Explaining a score does not show that changing the road helps.

Next: test it on real New Zealand roads

A retrospective pilot with a council or road authority.

1

Gather council data

Crash records, road details and traffic volumes.

2

Compare fairly

Does the shortlist beat how sites are chosen today?

3

Engineer review

Record where engineers agree or disagree with the list.

SDG 3.6 Fewer road deaths and injuries

SDG 11.2 Safe transport for everyone

An explainable shortlist for road safety checks

Next: test it on real New Zealand roads.

Live demo

Source code

Demo roadaccident-roshanar-aryal.streamlit.app
Code github.com/roshanaryal1/SafeRoute

Roshan Aryal

Thank you. Questions welcome.

Predicted-score concentration by shortlist size

Ranked by predicted practice score. Each shortlist includes the rows above it.

Shortlist	Examples	Score share	Concentration
Top 1%	5,177	2.2%	2.23×
Top 5%	25,887	9.9%	1.98×
Top 10%	**51,775**	**18.2%**	**1.82×**
Top 25%	129,438	39.3%	1.57×
All examples	517,754	100%	1.00×

Performance by road type, lighting and weather

Held-out 20%: about 104,000 examples. Overall R^2 0.8835, average error 0.044.

Split	Groups	Score variation explained (R^2)
Road type	Highway / Rural / Urban	88.4% / 88.5% / 88.1%
Lighting	**Daylight / Dim / Night**	**85.7% / 85.1% / 84.6%**
Weather	Clear / Foggy / Rainy	87.7% / 88.4% / 87.5%

Regional and demographic fairness could not be tested: the data has no such fields.